Musculoskeletal

Scoliosis



What: Sideways spinal curve with vertebral rotation. Adolescent idiopathic scoliosis often appears during a growth spurt.

Recognize: Uneven shoulders, waist or hips; rib prominence on the Adams forward-bend test. Often painless.

Diagnosis: Standing spinal X-rays measure the Cobb angle and guide follow-up.

Treatment: Depends on curve size, progression and remaining growth. Smaller curves are observed; bracing limits progression while growing; severe curves may need spinal fusion.

Typical thresholds: Observe below 25 degrees; brace about 25-45 degrees in a growing child; consider surgery above 45-50 degrees. Individual plans vary.

Nursing: Reinforce prescribed brace wear, check skin and fit, encourage activity allowed by the team, and support body image. Keep follow-up appointments. [1]

Torticollis



What: Tight/short sternocleidomastoid (SCM) muscle.

Signs: Head tilts toward the tight side; chin turns away. Limited neck motion; sometimes an SCM lump.

Assess: Neck motion, head shape and feeding; screen for associated DDH. Usually a clinical diagnosis.

Care: Early physical therapy, gentle taught stretches and varied awake positioning; supervised tummy time. Supine sleep.

Follow-up: Check symmetry and progress. New painful or acute torticollis needs evaluation for other causes. [2]

Plagiocephaly



What: Positional skull flattening from external pressure; sutures remain open.

Signs: Flat occiput; ear/forehead may shift forward on that side. Assess torticollis.

Diagnosis: Clinical exam; assess for craniosynostosis if atypical or worsening.

Care: Vary awake positioning, supervised tummy time and PT if needed. Always supine for sleep; no positioning pillows.

Treatment: Helmet for selected persistent severe cases after specialist review. [3]

Craniosynostosis



What: Premature fusion of one or more skull sutures.

Signs: Abnormal head shape; a firm suture ridge or altered head growth.

Care: Early craniofacial/neurosurgical referral; imaging as directed. Surgery often occurs in infancy. Monitor head growth and neurologic status; raised intracranial pressure is a risk. [4]

Clubfoot



What: Congenital rigid foot deformity: foot turns inward and down, with a high arch and inward heel. Diagnosed by examination.

Treatment: Early Ponseti stretching and serial casts; often Achilles tenotomy, then boots-and-bar bracing for several years.

Nursing: Check toes, circulation and skin; keep casts dry. Brace adherence prevents relapse. Report swelling, cool toes or persistent pain. [5,10,12]

Developmental Dysplasia of the Hip (DDH)



What: Shallow acetabulum with an unstable, subluxed or dislocated femoral head. Risks include breech position, family history and female sex.

Signs: Limited hip abduction, unequal leg length or later limp. Trained clinicians perform Barlow (provokes instability) and Ortolani (reduces a dislocated hip) maneuvers.

Diagnosis: Hip ultrasound in young infants; radiographs become useful as the femoral head ossifies.

Treatment: Pavlik harness in early infancy; reduction and hip-spica cast or surgery when indicated.

Nursing: Do not change prescribed strap settings. Check skin and leg movement/circulation, provide diaper/cast care and attend follow-up. Swaddle with hips free to flex and spread. [6]







Slipped Capital Femoral Epiphysis



What: Femoral head slips at the growth plate, usually during adolescence. Risks include obesity and endocrine disorders.

Signs: Limp, hip/groin/thigh or referred knee pain; outward-turned leg and reduced internal rotation.

Diagnosis: Hip/pelvis X-rays. Inability to bear weight indicates an unstable slip.

Priority: Stop weight bearing and obtain urgent orthopedic evaluation. Do not force hip movement.

Treatment: Surgical fixation; follow ordered activity limits and assess pain/neurovascular status. Watch for avascular necrosis and later arthritis. [7]

Legg-Calve-Perthes Disease



What: Temporary loss of femoral-head blood supply causes avascular necrosis, followed by repair. Usually ages 4-10, more often boys.

Signs: Limp with hip, thigh or knee pain; reduced abduction and internal rotation. Activity may worsen pain.

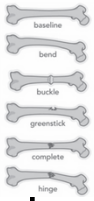
Diagnosis: Hip X-rays; follow imaging over the healing period.

Treatment: Maintain range of motion and contain the femoral head in the socket while it heals. Activity/weight-bearing limits, physical therapy, anti-inflammatory medication; selected casting, bracing or surgery.

Nursing: Support the prescribed mobility plan, avoid high-impact activity, reinforce follow-up and assess pain. Healing takes years; residual deformity can cause early arthritis. [8]

Neuromuscular Malformations

Fractures



Types: Buckle = compression; greenstick = incomplete break with bending; complete = through the bone. Growth-plate injury may affect future growth.

Signs: Pain, swelling, deformity and refusal to use the limb. Confirm with X-rays.

First care: Immobilize, manage pain and check distal color, warmth, capillary refill, pulses, sensation and movement before and after treatment.

Treatment: Splint/cast; reduction or fixation if displaced or unstable. Cover an open wound with a sterile dressing and obtain urgent care.

Follow-up: Recheck pain and neurovascular status, teach activity limits and keep healing/growth checks. [9,12]

Casting



Care: Support wet plaster with palms; keep cast dry and avoid pressure on edges/heels.

Routine swelling: Elevate as ordered and encourage finger/toe movement. Follow prescribed weight-bearing limits.

Never: Insert objects, powders or lotions; trim or alter the cast yourself.

Assess/report: Increasing pain, numbness, pale/cool digits, reduced motion, odor, drainage, fever or a tight/damaged cast. Check neurovascular status regularly. [10,12]

Traction

Purpose: Maintain alignment, immobilize the limb and reduce muscle spasm. Skin traction uses straps; skeletal traction uses bone pins.

Maintain: Prescribed line of pull and countertraction; ropes in pulleys and weights hanging freely. Do not change/remove weights without an order.



Nursing: Skin/pin-site care per protocol, neurovascular checks, pressure relief, pain control and immobility prevention. [11,12]

Complications

Compartment syndrome: Escalating pain despite analgesia, pain with passive stretch, tingling and tense swelling. Pulses may remain present; pulse loss/paralysis are late.

Action: Urgent surgical review. Keep limb at heart level; trained staff relieve constriction as directed. Fasciotomy may be required.

Also monitor: Pressure injury, infection, stiffness and muscle wasting with immobilization. [10,12]

Muscular Dystrophy



What: Genetic disorders causing progressive muscle weakness. Duchenne (DMD) is X-linked, mainly affects boys and involves absent/deficient dystrophin.

Signs: Early-childhood proximal weakness, frequent falls, waddling/toe-walking, enlarged calves and Gowers maneuver (hands push up thighs to stand).

Diagnosis: Elevated creatine kinase suggests muscle injury; genetic testing identifies the cause. Biopsy is sometimes needed.

Care: Prescribed disease-specific therapy/corticosteroids, stretching, PT and assistive devices; monitor cardiac and respiratory function.

Nursing: Prevent contractures and falls, protect nutrition/bone health, support independence and family/genetic counseling. [13,17]



Juvenile Idiopathic Arthritis







What: Immune-mediated inflammatory arthritis beginning before age 16, lasting at least 6 weeks after other causes are excluded.

Signs: Swollen, warm joints; morning stiffness and limp. Systemic forms may cause fever and rash.

Diagnosis: History/exam, inflammatory markers and selected imaging; no single confirmatory blood test.

Treatment: Rheumatology care; NSAIDs for symptoms, DMARDs such as methotrexate and/or biologics to control inflammation; selected corticosteroids.

Nursing: Balance rest with gentle motion and low-impact exercise (swimming). Warmth eases stiffness; cold can ease swelling. Monitor growth, function and medicine effects.

Eye care: Scheduled slit-lamp exams are essential because uveitis may be silent. [14-16]

Musculoskeletal concept maps: references

Blue notes complete the 14 original boxes. Original text, drawings and photographs are preserved.
Source numbers refer to the notes on pages 1-2. References checked November 3, 2026.

[1] AAOS: Idiopathic scoliosis

[2] Royal Children's Hospital: Congenital torticollis

[3] Royal Children's Hospital: Positional plagiocephaly

[4] CDC: Craniosynostosis

[5] AAOS: Clubfoot

[6] AAOS: Developmental dysplasia of the hip

[7] AAOS: Slipped capital femoral epiphysis

[8] AAOS: Perthes disease

[9] AAOS: Forearm fractures in children

[10] Royal Children's Hospital: Cast care

[11] Royal Children's Hospital: Skin traction

[12] Royal Children's Hospital: Neurovascular observations

[13] AAOS: Muscular dystrophy

[14] NIAMS: Juvenile idiopathic arthritis, diagnosis and treatment

[15] NIAMS: Juvenile idiopathic arthritis, symptoms

[16] ACR: JIA-associated uveitis guideline

[17] Muscular Dystrophy Association: Duchenne muscular dystrophy

Study notes accompany the course map; individual treatment plans and institutional protocols vary.